Sanchali Saha

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Skills Summary

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- **Languages**: JavaScript, TypeScript, Python, Java, C
- **Frontend**: HTML5, CSS3, React.js, Tailwind CSS
- **Backend & Database**: Node.js, Express.js, MongoDB, Firebase, Supabase, SQL
- **AI / Data Science**: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Core CS**: Object-Oriented Programming, Data Structures & Algorithms, DBMS, Operating Systems, Software Engineering
- **Tools**: Git, GitHub, Postman, Anaconda, Render, Vercel, Figma, Framer, Canva

Projects

- **NeatNode — AI-Powered CSV Data Cleaning Platform**
  - Engineered a full-stack data cleaning platform that automates column profiling, detects outliers using the IQR method, and applies median/mode imputation to handle missing values, reducing manual data preparation time.
  - Built a natural language query assistant using the OpenRouter LLM API and SQLite, with a local fast-path resolution layer to handle frequent queries without external API calls, reducing latency and costs.
  - Designed an interactive analytics dashboard for data quality metrics, profiling statistics, and cleaning summaries, with one-click export to CSV, JSON, SQL, and Python formats.
  - Deployed the React/Vite frontend on Vercel and Flask/Python backend on Render, resolving CORS configuration and build pipeline issues.
  - **Tech Stack:** React, Vite, Flask, Python, Pandas, SQLite, OpenRouter API
  - **GitHub:** Link — **Live:** Website
- **Agni-Drishti — AI-Enabled Geospatial System for Industrial Fire Detection**
  - Engineered a VIIRS satellite-data pipeline using DBSCAN clustering, 750m grid-cell profiling, Mann-Kendall trend analysis, and Temporal Concentration Index filtering to identify persistent industrial heat sources.
  - Built a two-branch classification system combining Global Energy Monitor matching with XGBoost-based thermal fingerprint analysis to classify Oil & Gas, Mining/Extraction, Metallurgical, and Other Industrial sources, achieving 81.9% accuracy and 75.3% macro F1 on an untouched test set.
  - Designed a GIS dashboard with seven intelligence modules for hotspot monitoring, event assessment, historical change and anomaly detection, and source investigation, featuring interactive maps, activity timelines, and thermal fingerprints.
  - Authored the system architecture document and pitch deck for Smart India Hackathon 2026 (SIH26162), covering the technical pipeline, feasibility, viability, and validated results.
  - **Tech Stack:** React, TypeScript, Leaflet, FastAPI, Python, PostGIS, XGBoost, pandas, GeoPandas, scikit-learn
  - **Live:** Website

Honors / Awards / Achievements

- **Winner — ShowcaseX (Best Use of Gemini), RCCIIT, Beleghata**: Built a civic issue reporting web app that helps citizens report potholes, garbage, broken streetlights, and other local problems, visualizing them on a city map for easier tracking and resolution.
- **Graphics Associate — GDG on Campus RCCIIT**: Designed graphics and visual content for events, social media campaigns, and community engagement initiatives.
- **Social Media Manager — SlashByte (Remote)**: Managed the company's LinkedIn presence by creating and publishing content to enhance brand visibility and community engagement.

Education

- **RCC Institute of Information Technology (RCCIIT)** — Kolkata, WB, India
  B.Tech in Information Technology — CGPA: 8.42 — *Aug 2024 – July 2028*